Evaluation runs: **chan** (reliable-channel-aneris)

generated 2026-11-05 from brain-data a1150b5a2 2026-11-05 14:21:32 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	882	conforms	0.1	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3	48	23800	conforms	0.7	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4	384	539540	conforms	7.0	4	1 h, 250 GB	1	1	brain03, 2026-11-02
5	3840	11217600	conforms	83.0	58	1 h, 250 GB	1	1	brain03, 2026-11-02
6	46080	221512004	conforms	2024.8	982	1 h, 250 GB	1	1	brain03, 2026-11-02
7	645120		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	8	704	362	violation	0.2	2	1 h, 250 GB	363	2897	brain03, 2026-11-02
3	48	20384	15064	violation	24.4	2	1 h, 250 GB	15065	723073	brain03, 2026-11-02
4	384	543060	462840	violation	1847.5	14	1 h, 250 GB	462841	177730561	brain03, 2026-11-02
5	3840			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 5 no-seqno bag, 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag correct, 7 correct bag, 7 bag no-seqno.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	8	882	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
3	48	23800	conforms	1.0	2	1 h, 250 GB	brain02, 2026-11-01
4	384	539540	conforms	8.6	6	1 h, 250 GB	brain02, 2026-11-01
5	3840	11217600	conforms	97.0	60	48 h, 250 GB	brain05, 2026-11-02
6	46080	221512004	conforms	2462.0	984	48 h, 250 GB	brain05, 2026-11-02
7	645120	4226823920	conforms	53486.0	19194	48 h, 250 GB	brain01, 2026-11-03

stateful: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	8	704	362	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-01
3	48	20384	15064	violation	1.1	2	1 h, 250 GB	brain02, 2026-11-01
4	384	543060	462840	violation	9.0	8	1 h, 250 GB	brain02, 2026-11-01
5	3840	14061696	12864192	violation	129.7	139	48 h, 250 GB	brain05, 2026-11-02
6	46080	362542466	344704646	violation	4067.0	2208	48 h, 250 GB	brain05, 2026-11-02
7				(48 h)			48 h, 250 GB	brain04, 2026-11-02
8				(48 h)			48 h, 250 GB	brain02, 2026-11-03

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 6 no-seqno bag, 7 correct bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag, 10 correct bag, 10 no-seqno bag, 11 no-seqno bag, 11 correct bag, 12 correct bag, 12 no-seqno bag, 13 no-seqno bag, 13 correct bag.

left out by the table rules (ok): 5 correct bag, 5 no-seqno bag, 6 correct bag.

left out by the table rules (timeout): 6 no-seqno bag, 7 correct bag.

skipped: a smaller size had reached its limit: 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

the tool ended with an error (exit-101): 8 bag correct.

left out by the table rules (invalid: foreign load): 9 bag no-seqno, 9 bag correct.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	23800	conforms	1.1	4	10 min, 16 GB	0.1	1.1	0.0	in-memory	brain05, 2026-11-02
4	384	539540	conforms	8.7	6	10 min, 16 GB	0.1	8.7	0.0	in-memory	brain05, 2026-11-02
5	3840	11217600	conforms	101.7	54	1 h, 250 GB	0.4	101.7	0.0	merge	brain05, 2026-11-03
6	46080	221512004	conforms	3464.8	978	1 h, 250 GB	1.0	3464.6	0.1	merge	brain05, 2026-11-03

offline: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	48	20384	87	violation	0.9	4	10 min, 16 GB	0.1	0.9	0.0	in-memory	brain05, 2026-11-02
4	384	543060	1092	violation	10.8	22	10 min, 16 GB	0.1	10.8	0.0	in-memory	brain05, 2026-11-02
5	3840	14061696	14413	violation	122.5	74	1 h, 250 GB	0.4	122.1	0.4	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 no-seqno bag, 6 bag correct, 7 bag correct, 7 correct bag, 7 bag no-seqno.

skipped: a smaller size had reached its limit: 6 no-seqno bag, 7 correct bag, 7 no-seqno bag, 8 correct bag, 8 no-seqno bag, 9 no-seqno bag, 9 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	11.1	2031	10 min, 250 GB	0.1	11.1	0.1	merge	brain01, 2026-11-03
3	48	23800	conforms	84.4	4566	10 min, 250 GB	0.1	84.3	0.1	merge	brain01, 2026-11-03

cross offline, Java implementation against the Rust specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	6.3	205	10 min, 250 GB	0.1	6.1	0.1	merge	brain01, 2026-11-03
3	48	20384	87	violation	42.7	4312	10 min, 250 GB	0.1	42.6	0.1	merge	brain01, 2026-11-03

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag correct, 7 bag no-seqno.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	882	conforms	4.1	171	10 min, 250 GB	4.0	0.1	0.0	merge	brain01, 2026-11-03
3	48	23800	conforms	4.4	168	10 min, 250 GB	4.3	1.1	0.1	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	704	6	violation	4.1	171	10 min, 250 GB	4.0	0.1	0.0	merge	brain01, 2026-11-03
3	48	20384	87	violation	4.4	168	10 min, 250 GB	4.3	0.9	0.1	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag correct, 7 bag no-seqno.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	882	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	48	23800	conforms	1.3	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	704	362	violation	0.1	4	1 h, 250 GB	11	brain01, 2026-11-02
3	48	20384	15064	violation	1.2	4	1 h, 250 GB	1	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag no-seqno, 5 bag correct, 6 bag no-seqno, 6 bag correct, 7 bag correct, 7 bag no-seqno.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	11.2	2031	10 min, 250 GB	11.1	0.1	0.1	merge	brain01, 2026-11-03
3	23800	23800	conforms	84.4	4566	10 min, 250 GB	84.3	1.1	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.2	205	10 min, 250 GB	6.1	0.1	0.1	merge	brain01, 2026-11-03
3	20384	20384	conforms	42.6	4312	10 min, 250 GB	42.6	0.9	0.1	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	11.1	2031	10 min, 250 GB	11.1	0.1	0.0	merge	brain01, 2026-11-03
3	23800	20384	87	violation	84.4	4566	10 min, 250 GB	84.3	0.9	0.1	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag no-seqno, 6 bag correct.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	11.3	2031	10 min, 250 GB	0.1	11.1	0.2	merge	brain01, 2026-11-03
3	23800	23800	conforms	84.4	4566	10 min, 250 GB	1.1	84.3	0.1	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.2	205	10 min, 250 GB	0.1	6.1	0.0	merge	brain01, 2026-11-03
3	20384	20384	conforms	42.8	4312	10 min, 250 GB	0.9	42.6	0.2	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	6.2	205	10 min, 250 GB	0.1	6.1	0.1	merge	brain01, 2026-11-03
3	23800	20384	87	violation	42.8	4312	10 min, 250 GB	1.1	42.6	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag no-seqno, 6 bag correct.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	8.5	256	10 min, 250 GB	8.5	0.1	0.0	merge	brain01, 2026-11-03
3	23800	23800	conforms	52.3	5010	10 min, 250 GB	52.0	1.1	0.3	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.0	234	10 min, 250 GB	5.9	0.1	0.0	merge	brain01, 2026-11-03
3	20384	20384	conforms	41.0	4364	10 min, 250 GB	40.9	0.9	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	8.6	256	10 min, 250 GB	8.5	0.1	0.1	merge	brain01, 2026-11-03
3	23800	20384	87	violation	52.1	5010	10 min, 250 GB	52.0	0.9	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag no-seqno, 6 bag correct.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	882	conforms	8.6	256	10 min, 250 GB	0.1	8.5	0.1	merge	brain01, 2026-11-03
3	23800	23800	conforms	52.1	5010	10 min, 250 GB	1.1	52.0	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	704	704	conforms	6.0	234	10 min, 250 GB	0.1	5.9	0.1	merge	brain01, 2026-11-03
3	20384	20384	conforms	40.9	4364	10 min, 250 GB	0.9	40.9	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	882	704	6	violation	5.9	234	10 min, 250 GB	0.1	5.9	0.0	merge	brain01, 2026-11-03
3	23800	20384	87	violation	41.1	4364	10 min, 250 GB	1.1	40.9	0.2	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 5 bag correct, 5 bag no-seqno, 6 bag correct.

planned but never scheduled (grid points beyond the ladder): 6 bag no-seqno.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	9.8	280	1 h, 250 GB		brain01, 2026-11-03
3	23800	23800	conforms	63.3	5055	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	9.1	236	1 h, 250 GB		brain01, 2026-11-03
3	20384	20384	conforms	56.8	4930	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	9.9	236	1 h, 250 GB	95	brain01, 2026-11-03
3	23800	20384	15064	violation	56.5	4658	1 h, 250 GB	142	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag no-seqno, 6 bag no-seqno, 6 bag correct.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	882	882	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	23800	23800	conforms	1.4	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	704	704	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02
3	20384	20384	conforms	1.4	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **no-seqno**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	882	704	362	violation	0.1	4	1 h, 250 GB	10	brain01, 2026-11-02
3	23800	20384	15064	violation	1.1	4	1 h, 250 GB	7	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). skipped: a smaller size had reached its limit: 5 bag correct, 5 bag no-seqno, 6 bag no-seqno, 6 bag correct.
planned but never scheduled (grid points beyond the ladder): 6 bag no-seqno.